ACI 318-19 Example 001

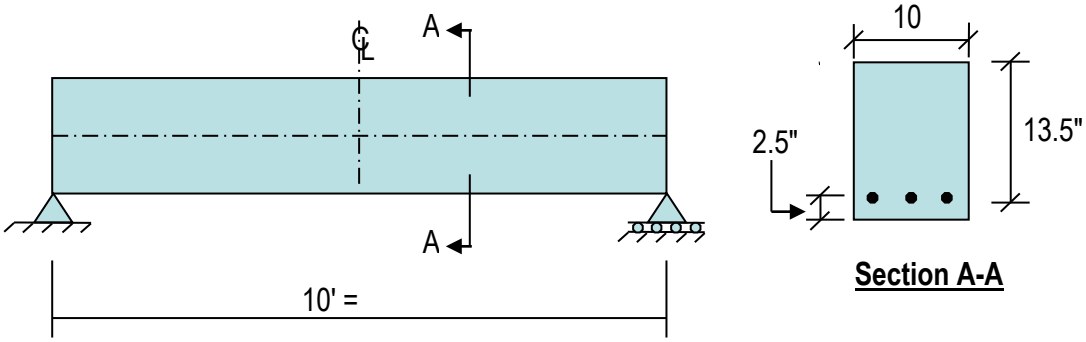
SHEAR AND FLEXURAL REINFORCEMENT DESIGN OF A SINGLY REINFORCED RECTANGLE

EXAMPLE DESCRIPTION

The flexural and shear design of a rectangular concrete beam is calculated in this example.

A simply supported beam is subjected to an ultimate uniform load of 9.736 k/ft. This example is tested using the ACI 318-19 concrete design code. The flexural and shear reinforcing computed is compared with independent hand calculated results.

GEOMETRY, PROPERTIES AND LOADING



Material Properties		Section Properties			Design Properties
E =	3600 k/in^2	d	=	13.5 in	f'$_c$ = 4 k/in^2
v =	0.2	b	=	10.0 in	f_y = 60 k/in^2
G =	1500 k/in^2	I	=	3,413 in^4	

TECHNICAL FEATURES TESTED

- Calculation of Flexural reinforcement, A_s
- Enforcement of Minimum tension reinforcement, $A_{s,min}$
- Calculation of Shear reinforcement, A_v
- Enforcement of Minimum shear reinforcing, $A_{v,min}$

RESULTS COMPARISON

Independent results are hand calculated based on the equivalent rectangular stress distribution described in Example 6.1 in Notes on ACI 318-19 Building Code.

Output Parameter	ETABS	Independent	Percent Difference
Design Moment, M_u (k-in)	1460.4	1460.4	0.00%
Tension Reinf, A_s (in^2)	2.37	2.37	0.00%
Design Shear Force, V_u	37.73	37.73	0.00%
Shear Reinf, A_v/s (in^2/in)	0.041	0.041	0.00%

COMPUTER FILE: ACI 318-19 Ex001

CONCLUSION

The computed results show an exact match for the flexural and the shear reinforcing.

HAND CALCULATION

Flexural Design

The following quantities are computed for all the load combinations:

$\varphi = 0.9,\ A_g = 160$ sq-in

$$A_{s,min} = \frac{200}{f_y} b_w d = 0.450 \text{ sq-in (Govern)}$$

$$= \frac{3\sqrt{f_c'}}{f_y} b_w d = 0.427 \text{ sq-in}$$

$$\beta_1 = 0.85 - 0.05\left(\frac{f_c' - 4000}{1000}\right) = 0.85$$

$$c_{max} = \frac{0.003}{0.003 + \left(0.003 + f_y / E\right)} d = 5.019 \text{ in}$$

$$a_{max} = \beta_1 c_{max} = 4.266 \text{ in}$$

Combo1

$$w_u = (1.2w_d + 1.6w_l) = 9.736 \text{ k/ft}$$

$$M_u = \frac{w_u l^2}{8} = 9.736 \bullet 10^2/8 = 121.7 \text{ k-ft} = 1460.4 \text{ k-in}$$

The depth of the compression block is given by:

$$a = d - \sqrt{d^2 - \frac{2|M_u|}{0.85 f_c' \varphi b}} = 4.183 \text{ in } (a < a_{max})$$

The area of tensile steel reinforcement is given by:

$$A_s = \frac{M_u}{\varphi f_y \left(d - \frac{a}{2}\right)} = \frac{1460.4}{0.9 \bullet 60 \bullet (13.5 - 4.183/2)} \text{ in}$$

$$A_s = 2.37 \text{ sq-in}$$

Shear Design

The following quantities are computed for all of the load combinations:

$\varphi = 0.75$

If the beam is provided with the minimum shear rebar, i.e. $A_v \geq A_{v,\min}$, the concrete shear capacity is given by:

$$V_c = \max \begin{cases} \left(2\lambda \sqrt{f'_c} + \dfrac{N_u}{6A_g} \right) A_{cv}, & \text{and} \\ \left(8\lambda (\rho_w)^{1/3} \sqrt{f'_c} + \dfrac{N_u}{6A_g} \right) A_{cv}. \end{cases}$$

If the beam is not provided with the minimum shear rebar, i.e. $A_v < A_{v,\min}$, the concrete shear capacity is given by:

$$V_c = \left(8\lambda_s \lambda (\rho_w)^{1/3} \sqrt{f'_c} + \frac{N_u}{6A_g} \right) A_{cv}$$

Check the limit of the following parameters:

$$\lambda_s = \sqrt{\frac{2}{1+\dfrac{d}{10}}} = \sqrt{\frac{2}{1+\dfrac{13.5}{10}}} = 0.923 \leq 1.00$$

$\lambda = 1$ (normal weight concrete)

$$\rho_w = \frac{A_s}{b\,d}$$

$\dfrac{N_u}{6A_g} = 0$ (throughout the length of the beam)

$A_{cv} = b\,d = 10 \bullet 13.5 = 135$ sq-in

$\sqrt{f'_c}$ = 63.246 psi < 100 psi

$\phi 5\lambda \sqrt{f'_c}\, b\,d$ = 32.018 k

Given V_u, V_c and V_{max}, the required shear reinforcement in area/unit length for any load combination is calculated as follows:

If $V_u \leq \varphi\,(V_c/2)$,

$$\frac{A_v}{s} = 0,$$

else if $\varphi\,(V_c/2) < V_u \le \varphi\, V_{max}$

$$\frac{A_v}{s} = \frac{(V_u - \phi V_c)}{\phi\, f_{ys}\, d} \ge \left(\frac{A_v}{s}\right)_{min}$$

where:

$$\left(\frac{A_v}{s}\right)_{min} = \max\left\{50\left(\frac{b_w}{f_{yt}}\right),\ 0.75\left(\frac{b_w}{f_{yt}}\right)\sqrt{f'_c}\right\}$$

$$\left(\frac{A_v}{s}\right)_{min} = 0 \ \text{(if } V_u \le \phi V_c \text{, and } h \le 10 \text{ in for rectangular beam)}$$

else if $V_u > \varphi\, V_{max}$,

a failure condition is declared.

Combo1

The shear rebar is calculated at a distance d from the end (second station).

V_u = 9.736•(5-13.5/12) = 37.727 k

M_u = 583.25 kip-in

A_s = 0.847 sq-in (from ETABS)

$$\rho_w = \frac{A_s}{b\,d} = 0.00627$$

$$\lambda_s = 0.923$$

$$\lambda = 1$$

$$\frac{N_u}{6A_g} = 0$$

$$A_{cv} = b\,d = 10 \bullet 13.5 = 135 \text{ sq-in}$$

$\sqrt{f'_c}$ = 63.246 psi

Assume that the minimum shear rebar is not needed: i.e. $A_v < A_{v,\min}$:

$$\phi V_c = \phi\left(8\lambda_s\lambda(\rho_w)^{1/3}\sqrt{f'_c} + \frac{N_u}{6A_g}\right)A_{cv} = 8.716 \text{ k}$$

$V_u = 37.727 \text{ kip} > \phi V_c$ (minimum shear rebar is needed)

Assume that the minimum shear rebar is needed: i.e. $A_v \geq A_{v,\min}$:

$$V_c = \max\begin{cases}\left(2\lambda\sqrt{f'_c} + \frac{N_u}{6A_g}\right)A_{cv}, & \text{and} \\ \left(8\lambda(\rho_w)^{1/3}\sqrt{f'_c} + \frac{N_u}{6A_g}\right)A_{cv}.\end{cases}$$

$$V_c = \max\begin{cases}(123.491psi)\,A_{cv} \\ (93.317 \text{ psi})\,A_{cv}\end{cases}$$

The concrete shear capacity is given by:

$\varphi V_c = 12.807 \text{ k} < \phi 5\lambda\sqrt{f'_c}\,bd = 32.018 \text{ k}$

The maximum shear that can be carried by reinforcement is given by:

$\varphi V_s = \varphi 8\sqrt{f'_c}\,bd = 51.229 \text{ k}$

The following limits are required in the determination of the reinforcement:

$(\varphi V_c/2) = 6.4035 \text{ k}$

$(\varphi V_c/2) = 12.807 \text{ k}$

$\varphi V_{max} = \varphi V_c + \varphi V_s = 64.036 \text{ k}$

Shear rebar calculation:

$$\phi(V_c/2) = 6.4035k \leq V_u = 37.727k \leq \phi V_{\max} = 64.036k$$

$V_u = 37.727k > \phi V_c = 12.807k$ (exception clause 9.6.3.1 does not apply)

$$\left(\frac{A_v}{s}\right)_{\min} = \max\left\{50\left(\frac{10}{60,000}\right),\ 0.75\left(\frac{10}{60,000}\right)\sqrt{4,000}\right\}$$

$$\left(\frac{A_v}{s}\right)_{\min} = \max\left\{0.0083,\ 0.0079\right\} = 0.0083\frac{in^2}{in}$$

$$\frac{A_v}{s} = \frac{(V_u - \phi V_c)}{\phi f_{ys} d} = 0.041\frac{in^2}{in} = 0.492\frac{in^2}{ft}$$